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ORIGINAL ARTICLE

Facial one-pot synthesis of D_{3h} symmetric bicyclocalix[2]arene[2]triazines and their layered comb self-assembly

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Abstract A number of D_{3h} symmetric bicyclocalix[2]arene[2]triazine core compounds were synthesized *via* a general and good-yielding (43–48% yield) facile protocol starting from cyanuric halides, phloroglucinol and K_2CO_3 under very mild reaction conditions. These cage-like compounds are tolerante with different reaction conditions and can be derived with other functional groups in high yields. The X-ray crystal structures show these compounds have slightly distorted D_{3h} symmetric structures. Due to the unique molecular topological structure, bicyclocalix[2]arene[2]triazine molecules form unique layered comb networks when hydrogen bond groups exist (such as CO_2H, $B(OH)_2$), which represent a new kind of building block unit for supramolecular architectures.
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1. Introduction

Supramolecular self-assemblies, such as supramolecular cage [1] supramolecular plate [2] MOF [3] HOF [4] etc., have received great interests in the past few decades. The topological structure of the highly symmetric building-block molecule decided the architecture of the following supramolecular self-assemblies [1d,3,5]. Generally speaking, new geometric building block molecules usually bring new supramolecular architectures, and the function groups decide the way the building block molecules join each other. Most of the building blocks used at present are 2D planar molecules, building block molecules with 3D steric structures are rare.

Heteracalixaromatics are emerging supramolecular host molecule, due to their easy availability, rich structures and conformations, and powerful recognition properties [6]. They have varied 3D structures and functions and can be used as new type of building block for supramolecular architecture [7]. The cage-like heteracalixaromatics bearing enclosed cavities are particularly intriguing. Cage structure has rigid and stable configuration, it can be the ideal 3D building block unit for supramolecular architecture. The enclosed cavity of the cage also can work as a molecular flask and trap specific guest

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molecules [8] stabilize the trapped species [9] and endow intrinsic porosity [10].

Up to now, the reported cage-like molecules still are few, their application in the construction of supramolecular architecture is very limited. The preparation of metal-coordinated supramolecular cages and dynamic covalent bonded cages are convenient and efficient to some extent [1,8]. But these cages usually lack stability, diversity of function groups, which limit the application of these cage molecules in supramolecular self-assemblies. The chemical stable organic cages, especially the heteracalixaromatic cages, usually were synthesized *via* multistep fragment coupling strategies, which is time costly and low efficient [8b,8c,11]. Katz et al. reported one case for the facile synthesis of heteracalixaromatic cage, but it can't adopt different function groups [12].

Recently, Wang et al. reported a cage-like heteracalixaromatic compound, bicyclo(tetraoxacalix[2]arene[2]triazine), synthesized from phloroglucinol and cyanuric chloride for two steps [13]. This molecule adopts a D_3h symmetric rigid conformation, it can be derived with nucleophilic groups such as NR_2, OR, SR. The starting materials for this cage molecule are very cheap, and very mild conditions are needed for the reaction. This molecule represents a new class of flexible scaffolds for molecular and supramolecular architecture design. However, the triazine ring is highly electron deficient, only tolerant with very limited kinds of function groups (generally electron-donating groups). By introduction of a conjugated benzene ring to the triazine ring, here we developed the heteracalixaromatic cage into a universal D_3h symmetric scaffold.

2. Experimental

2.1. General information

^{1}H and ^{13}C NMR spectra were recorded on a Bruker Avance III 300M spectrometer. Chemical shifts are reported in ppm with tetramethylsilane or the residual solvent resonance as an internal standard. Melting points are uncorrected. Elemental analysis, mass spectrometry, TGA/DSC and single crystal X-ray diffraction analysis were performed on a Rigaku ST Saturn 724+. Solvents were dried according to standard procedures prior to use. All other major chemicals were obtained from commercial sources and used without further purification.

2.2. The single crystal X-ray diffraction

Crystals of **2c**-**2e** were taken from the mother liquor and dried on a filter paper, one single crystal with proper size was loaded on the top of a glass capillary for X-ray testing. The crystal of **2f** was weathering quickly in contact with the air, single crystal of **2f** was sealed in a glass capillary with mineral oil for test.

Intensity data were collected on a Rigaku ST Saturn 724+ diffractometer equipped with a Cu target X-ray sources and an CCD detector. Crystals were cooled to 100 K (unless specified otherwise in the crystallographic data) under N_2 flow from an Oxford Cryosystems Cryostat. Data reduction was performed by a standard procedure using SAINT and empirical corrections were performed, where necessary, using SADABS. Crystal structures were solved using SHELXS-2013 by direct methods and all ordered non-hydrogen atoms were refined anisotropically by means of fullmatrix least-squares calculations on F^2 by SHELXL-2013.

2.3. Synthesis of **3a**, **3b**, **3c** [14]

Cyanuric chloride(9.3 g, 50 mmol), anhydrous $AlCl_3$ (8.1 g, 60 mmol) and chlorobenzene (70 ml, or bromobenzene 150 °C, toluene r.t.) were added to a 250 ml single neck flask, the mixture was refluxed for 12 h until no more HCl gas released. Then, the mixture was cooled down and quenched by ice, the product was extracted by CH_2Cl_2 (60 ml*3). The extract was boiled with charcoal and evaporated by a rotating vacuum evaporator. The residue was recrystallized by the mixture of CH_2Cl_2 and petroleum ether to afford faint yellow crystal products.

3a (9.1 g, 72%): mp 148 °C; MS (ESI) *m*/*z* (%) 255.9 [M+H$^+$](1 0 0), 261.9 (95); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.47 (d, J = 8.6 Hz, 2H, Ar(**CH**)), 7.53 (d, J = 8.6 Hz, 2H Ar (**CH**)); ^{13}C NMR (75 MHz, $CDCl_3$) δ 173.8 (Tr(**C**)), 172.1 (Tr(**C**)), 141.4(Ar(**C**)), 131.2(Ar(**CH**)), 130.4(Ar(**C**)), 129.5(Ar (**CH**)).

3b (10.2 g, 67%): mp 165 °C; MS (ESI) *m*/*z* (%) 305.9 [M+H$^+$](1 0 0), 303.9 (62), 307.9(38); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.39 (d, J = 8.6 Hz, 2H Ar(**CH**)), 7.33 (d, J = 8.6 Hz, 2H Ar(**CH**)); ^{13}C NMR (75 MHz, $CDCl_3$) δ 174.0(Tr (**C**)), 172.2(Tr(**C**)), 132.5(Ar(**CH**)), 131.5(Ar(**C**)), 131.2(Ar (**CH**)), 130.3(Ar(**C**)).

3c (9.9 g, 83%): mp 146 °C; MS (ESI) *m*/*z* (%) 240.0 [M+H$^+$](1 0 0), 242.0 (63); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.39 (d, J = 8.4 Hz, 6H Ar(**CH**)), 7.33 (d, J = 8.4 Hz, 6H Ar (**CH**)), 2.46 (s, 3H, C**H3**); ^{13}C NMR (75 MHz, $CDCl_3$) δ 174.8(Tr(**C**)), 171.9(Tr(**C**)), 146.0(Ar(**C**)), 130.0(Ar(**CH**)), 130.0(Ar(**C**)), 129.9(Ar(**CH**)), 21.9(**C**H_3).

2.4. Synthesis of **2a**, **2b**, **2c**

3a (1.56 g, 6 mmol, 1.82 g for **3b**, 1.44 g for **3c**), phloroglucinol 0.25 g(2mmol), gridded K_2CO_3 (0.84 g, 6 mmol) and 50 ml acetone were added to a 50 ml single neck flask. The mixture was stirred at room temperature for 4 h, then another portion of phloroglucinol 0.25 g(2mmol), gridded K_2CO_3 (0.84 g, 6 mmol) and 50 ml acetone was added and refluxed for 2 h. After filtration, the filtrate was evaporated and the residue was recrystallized by the mixture of CH_2Cl_2 and petroleum ether to afford faint yellow crystal products.

2a (0.67 g, 46%): mp > 300 °C; MS (MALDI-TOF) *m*/*z* (%)814.0 [M+H$^+$](1 0 0), 816.0 (64),815.0 (41); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.54 (d, J = 8.4 Hz, 6H Ar(**CH**)), 7.55 (d, J = 8.4 Hz, 6H Ar(**CH**)), 6.78 (s, 6H, Ph(**CH**)); ^{13}C NMR (75 MHz, $CDCl_3$) δ 175.9(Tr(**C**)), 172.7(Tr(**C**)), 152.9 (Ph(**C**)), 139.8(Ar(**C**)), 132.0(Ar(**C**)), 130.2(Ar(**CH**)), 128.7(Ar (**CH**)), 114.7(Ph(**CH**)); IR (KBr) ν 2955, 1565, 1530, 1373, 1162, 1085, 1002, 815 cm^{-1}. Anal. Calcd. for $C_{39}H_{18}Cl_3N_9O_6$: C, 57.48; H, 2.23; N, 15.47. Found: C, 57.56; H, 2.31; N, 15.27.

2b (0.74 g, 43%): mp > 300 °C; MS (MALDI-TOF) *m*/*z* (%) 947.9 [M+H$^+$] (1 0 0), 949.9 (98), 950.9 (49); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.45 (d, J = 8.4 Hz, 6H, Ar(**CH**)), 7.71 (d, J = 8.4 Hz, 6H, Ar(**CH**)), 6.77 (s, 6H, Ph(**CH**)); ^{13}C

NMR (75 MHz, $CDCl_3$) δ 176.5(Tr(**C**)), 173.2(Tr(**C**)), 153.4 (Ph(**C**)), 132.9(Ar(**C**)), 132.2(Ar(**CH**)), 130.8(Ar(**CH**)), 129.1 (Ar(**C**)), 115.2(Ph(**CH**)); IR (KBr) ν 2955, 1566, 1533, 1372, 1161, 1088, 1002, 813 cm^{-1}. Anal. Calcd. for $C_{39}H_{18}Br_3N_9O_6$: C, 49.39; H, 1.91; N, 13.29. Found: C, 49.47; H, 2.10; N, 13.15.

2c (0.64 g, 48%): mp > 300 °C; MS (MALDI-TOF) *m/z* (%) 754.2 [M + H^+] (1 0 0), 755.2 (48), 756.2 (10); ^1H NMR (300 MHz, $CDCl_3$) δ 8.46 (d, *J* = 8.4 Hz, 6H, Ar(**CH**)), 7.34 (d, *J* = 8.4 Hz, 6H, Ar(**CH**)), 6.74 (s, 6H, Ph(**CH**)), 2.46 (s, 9H,C**H3**); ^{13}C NMR (75 MHz, $CDCl_3$) δ 177.2(Tr(**C**)), 173.1 (Tr(**C**)), 153.4(Ph(**C**)), 144.8(Ar(**C**)), 131.4(Ar(**C**)), 129.6(Ar (**CH**)), 129.5(Ar(**CH**)), 115.2(Ph(**CH**)), 21.8(**C**H_3); IR (KBr) ν 2973, 1565, 1531, 1373, 1160, 1085, 1002, 814 cm^{-1}. Anal. Calcd. for $C_{42}H_{27}N_9O_6$: C, 66.93; H, 3.61; N, 16.73; Found: C, 67.14; H, 3.73; N, 16.61.

2.5. Synthesis of **2d**:[15]

$Pd_2(dba)_3$mg (107 mg) and 20% PCy_3 in toluene (800 mg) were added to 20 ml anhydrous dioxane in a 50 ml schlenk flask under argon. After stirring for 20 min, 2a 1.3 g (1.6 mmol), bis(pinacolato)diboron (1.35 g, 5.2 mmol, 3.3 equiv.) and KOAc (1.2 g, 10.4 mmol, 6.6 equiv.) were added, the mixture was stirred at 80 °C for 3 h under argon water was added to the reaction mixture to quench the reaction. After evaporation, the residue was extracted CH_2Cl_2 and dried over Na_2SO_4. Flash chromatograph with ethyl acetate, petroleum ether = 1 :10–1:5 as the mobile phase afforded white solid products **2d**. **2d** was recrystallized by the mixture of acetone and hexane to give a single crystal.

2e (1.20 g, 73%): mp > 300 °C; MS (MALDI-TOF) *m/z* (%) 1090.4[M + H^+] (1 0 0), 1089.4 (50), 1091.4 (44); ^1H NMR (300 MHz, $CDCl_3$) δ 8.59 (d, *J* = 8.2 Hz, 6H, Ar (**CH**)), 8.01 (d, *J* = 8.2 Hz, 6H, Ar(**CH**)), 6.79 (s, 6H, Ph (**CH**)), 1.41 (s, 36H, C**H3**); ^{13}C NMR (75 MHz, $CDCl_3$) δ 176.8(Tr(**C**)), 172.8(Tr(**C**)), 152.9(Ph(**CH**)), 135.7(Ar(**C**)), 134.5(Ar(**CH**)), 127.9(Ar(**CH**)), 114.7(Ph(**CH**)), 83.7(**C**OMe_2), 24.4(**C**H_3); IR (KBr) ν 2979, 1569, 1528, 1431, 1357, 1166, 1111, 1085, 1002, 813 cm^{-1}. Anal. Calcd. for $C_{57}H_{54}B_3N_9O_{12}$: C, 62.84; H, 5.00; N, 11.57. Found: C, 62.57; H, 5.14; N, 11.21.

Synthesis of **2e**: [16] To a solution of the **2d** (1.09 g, 1.0 mmol) in acetone (30 mL) were added $NaIO_4$ (1.28 g, 6.0 mmol), NH_4OAc (460 mg, 6.0 mmol), and water (15 mL), and the mixture was stirred for 24 h. Acetone was evaporated, and the mixture was extracted with ethyl acetate, the extract was dried over Na_2SO_4, and the solvent was evaporated and dried in vacuo to afford **2e** as a white solid. **2e** was recrystallized by the mixture of acetone, ethyl acetate and hexane to give a single crystal.

2e (0.90 g, 98%): mp > 300 °C; MS (MALDI-TOF) *m/z* (%)844.2 [M + H^+] (1 0 0), 843.2 (74), 845.2 (42); ^1H NMR (300 MHz, DMSO d_6) δ 8.42 (d, *J* = 8.2 Hz, 6H, Ar(**CH**)), 8.30 (s, 6H, B(O**H**)$_2$), 8.00 (d, *J* = 8.2 Hz, 6H, Ar(**CH**)), 7.10 (s, 6H, Ph(**CH**)); ^{13}C NMR (75 MHz, DMSO d_6) δ 177.6(Tr (**C**)), 174.3(Tr(**C**)), 154.6(Ph(**C**)), 137.0 (Ar(**C**)), 135.3(Ar (**CH**)), 128.8 (Ar(**CH**)), 116.2(Ph(**CH**)); IR (KBr) ν 3404, 1565, 1524, 1428, 1373, 1160, 1084, 1004, 816 cm^{-1}. Anal. Calcd. for $C_{39}H_{24}B_3N_9O_{12}\cdot C_3H_6O\cdot C_4H_8O_2$: C, 55.85; H, 3.87; N, 12.74; Found: C, 55.68; H, 4.12; N, 12.38.

2.6. Synthesis of **2f**: [17]

CrO_3 (120 mg, 1.2 mmol), H_5IO_6 (320 mg, 8.4 mmol) were added to 40 ml acetonitrile in a 100 ml single neck flask. After stirring for a while, the solid was dissolved, **2c** (0.30 g, 0.4 mmol) dissolved in 10 ml acetonitrile/dichloromethane (1:1) was added to the flask dropwise in 15 min, and a yellow precipitate was generated upon addition. The mixture was stirred for another 10 h and evaporated. The residue was washed with water and extracted with ethyl acetate, the extract was dried over Na_2SO_4 and evaporated to afford 0.29 g off-white crude product with a yield of 90%. The crude product was recrystallized by the mixture of THF and acetonitrile to afford crystal product **2f**.

2f (0.30 g, 89%): mp > 300 °C; MS (MALDI-TOF) *m/z* (%) 844.1 [M + H^+] (1 0 0), 845.1 (48), 846.1 (13); ^1H NMR (300 MHz, DMSO d_6) δ 13.36(s, 6H, COO**H**), 8.56 (d, *J* = 8. 4 Hz, 6H, Ar(**CH**)), 8.16 (d, *J* = 8.4 Hz, 6H, Ar(**CH**)), 7.12 (s, 6H, Ph(**CH**)); ^{13}C NMR (75 MHz, DMSO d_6) δ 175.5(Tr (**C**), 173.3(Tr(**C**), 167.1 (**C**OOH), 153.4(Ph(**C**)), 138.1(Ar(**C**)), 135.5(Ar(**CH**)), 130.4(Ar(**CH**)), 129.4(Ar(**C**)), 115.8(Ph(**CH**)); IR (KBr) ν 3442, 1563, 1526, 1424, 1370, 1162, 1082, 1001, 814 cm^{-1}. Anal. Calcd. for $C_{42}H_{21}N_9O_{12}$: C, 59.79; H, 2.51; N, 14.94; Found: C, 59.50; H, 2.62; N, 14.77.

3. Results and discussion

We initiated the synthesis of functionalized D_3h symmetric cages by stepwise fragment coupling strategy (Fig. 1). The scaffold bicyclocalix[2]arene[2]triazine cage **1** was synthesized *via* two steps, then the benzene ring was introduced to the cage molecule by Suzuki coupling.

At first, we studied the Suzuki coupling reaction of cage **1** and 4-chlorophenylboronic acid. When THF was used as the solvent, 2.5 mol% $Pd(PPh_3)_4$ as the catalyst, 2 M aq·Na_2CO_3 as the base, no product was produced (Table 1). After reaction condition optimization, a mixture of toluene: THF = 10: 1 was used as the solvent, and 3 equiv. of 0.8 M aqueous Na_2CO_3 was used as the base, the cage compound **2** (X = Cl) can be synthesized with a yield of 78%.

With the same reaction conditions, **1** can react with 4-bromophenylboronic acid. However, no cage compound **2** (X = Br) can be separated as the product, due to the high reactivity of the Br group, only chain propagation polymer products were found, the substrate scope for the reaction is limited. Meanwhile, the synthesis of **1** is costly and is of low efficiency due to the fragment strategy needing highly diluted conditions. It's really painful to synthesize **2** *via* this strategy.

Inspired by the one-pot synthesis of oxygen bridged calix[2] fluoroarene[2]triazine.[18] Here we developed a one-pot synthesis protocol for cage compounds **2**, to make the bicyclocalix[2]arene[2]triazine as an easily available and universal D_3h symmetric scaffold. At first, the functionalized benzene ring was introduced to cyanuric chloride to produce **3** with high yields *via* Friedel–Crafts reaction (Fig. 2). Then the bichloride triazine unit can react with phloroglucinol to form the thermodynamic stable product **2**. We started the synthesis of **2** with **3a**, **3a** readily condensed with phloroglucinol in the presence of K_2CO_3 as a base to convert to the functionalized cage **2a**. 3.0 equiv. of **3a** and 2.05 equiv. of phloroglucinol

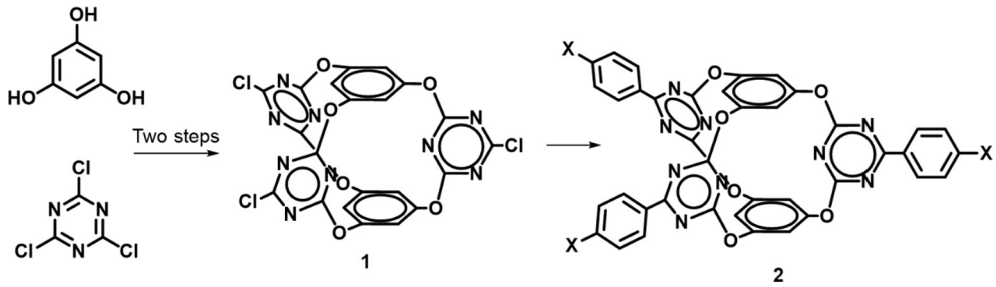

Fig. 1 Synthesis of D_3h symmetric cages via stepwise fragment coupling approach.

Table 1 Synthesis of 2 with stepwise strategy.

Solvent	Conditions	Yield
THF	Reflux, 6 h, 2.5%M $Pd(PPh_3)_4$, 2 M aq. Na_2CO_3 3 equiv.	No yield
Dioxane	Reflux, 6 h, 2.5%M $Pd(PPh_3)_4$, 2 M aq. Na_2CO_3, 3 equiv.	No yield
Toluene	80 °C, 6 h, 2.5%M $Pd(PPh_3)_4$, 2 M aq. Na_2CO_3, 3 equiv.	36%
Toluene: THF = 5:1	60 °C, 4 h, 2.5%M $Pd(PPh_3)_4$, 0.8 M aq. Na_2CO_3, 3 equiv.	69%
Toluene: THF = 10:1	60 °C, 4 h, 2.5%M $Pd(PPh_3)_4$, 0.8 M aq. Na_2CO_3, 3 equiv.	78%

refluxed with K_2CO_3 (Table 2, entry 1), afford **2a** with a yield of 23%; While the phloroglucinol was added to the reaction as two half, at first reacting under room temperature for 4 h then reflux for 2 h, the isolated yield of **2a** can be improved to 46% (Table 2, entry 2). **3b**, **3c** behave the same as **3a** in the reaction with phloroglucinol to produce functionalized cage compounds **2b**, **2c**, as well as similar yield (Table 2).

Further study found that the high dilution is not required to prevent polymerization of the starting materials, ambient atmosphere or moisture also doesn't affect the isolated yield of **2**. The synthesis procedure is simple and easy to handle.

With Cl, Br, CH_3 as a direction group, different kinds of function groups can be introduced to **2** by catalyzed coupling or general organic reaction (Fig. 3), for example, $B(OR)_2$, $B(OH)_2$, CO_2H groups can be introduced with high yields (Table 3). Borate or boric acid group functionalized **2** can be obtained by metal catalyzed coupling of **2a**/**b** with bis(pinacolato)diboron, carboxylic group functionalized **2** can be obtained by oxidation of **2c**. That identified that the calixarene cage scaffold is tolerant with different reaction conditions.

All of these macrocyclic products as synthesized are crystalline compounds, and their structures were characterized by NMR, EA, IR and MS. To understand the conformation structures of **2**, single crystals of **2c**, **2e**, **2d** and **2f** were cultivated and determined by X-ray.

X-ray diffraction results show that these compounds have quasi D_3h symmetric propeller shaped structure. The arms may slightly twist due to hydrogen bond interaction, π-π interaction or steric effect. As depicted in Fig. 4a, bicyclocalix[2]arene[2]triazines **2c** adopts quasi-D_3h symmetric conformation (Fig. S1). The average length of the bridging C—C bond ($d_{C(benzene)-C(triazine)}$) is 1.47 Å, which is between the C—C bond in benzene ring (1.39 Å) and the benzene methyl group ($d_{C(benzene)-C(CH3)}$, 1.52 Å). That shows that the benzene ring conjugates with the triazine ring very well, these two aromatic rings tend to stay in the same plane and present as three long arms of a propeller. However, due to the steric effect and π-π stacking, the arms were twisted to some extent. The substituent groups on the benzene ring didn't affect the conformation of the cage.

As observed, when the X group was changed to pinacolatoboron, the structure of cage molecule **2d** is similar to that of **2c** (Fig. 4b), but more twisted due to the bulky pinacolatoboron groups (Fig. S2).

When hydrogen bond groups were introduced to the benzene ring, the situation is different. As shown in Fig. 4c, the triboric acid **2e** adopts a twisted D_3h symmetric cage conformation. The average length of the bridging C—C bond ($d_{C(benzene)-C(triazine)}$) is 1.48 Å. This structure is rigid, one arm of the molecule is straight forward, but the other two arms are bended due to the intermolecular hydrogen bond interaction (Fig. S3). The $B(OH)_2$ groups of one molecule interact with $B(OH)_2$ groups of two other molecules in the neighborhood to form comb-like structure, the V-shape cleft of the bicyclocalix[2]arene[2]triazines was forced to be straight and nearly disappeared.

Interestingly, we found cage molecules **2e** self-assemble in a distinctive way in the solid state (Fig. 5a). Hydrogen bond directed comb-like **2e** assembly interlocks with another one *via* π-π stacking to form ladder-like structure. The nano ladders further pile up to form 3D honey-comb layer, which is around 6.6 Å thick. In the layer, four **2e** molecules interact with each other to form a quasi-square pore, the size of the pore is 21.0 Å * 21.0 Å * 6.6 Å. Due to the honey-comb layers staggered on each other, no channel structure can be observed in the crystal of **2e**, meanwhile, the nano pore is occupied by the solvent molecules co-crystallized. That coincides with the BET test result, which shows a surface area only less than 10 m^2/g for **2e** crystal.

TGA/DSC analysis of **2e** crystal shows this assembly has high thermal stability (Fig. S6). When the sample was heated up, it slowly lost the weight with time, the sample weight decreased near linearly with the increase in the temperature up to 200 °C, which most possibly due to the dissociation of solvent molecules in the shallow layers of the hydrogen bond frameworks (co-crystallized solvent take cal. 15 wt%). In 200–400 °C region, the sample was nearly stable and lost less than 3% weight. The compound began to decompose when the temperature was over 400 °C.

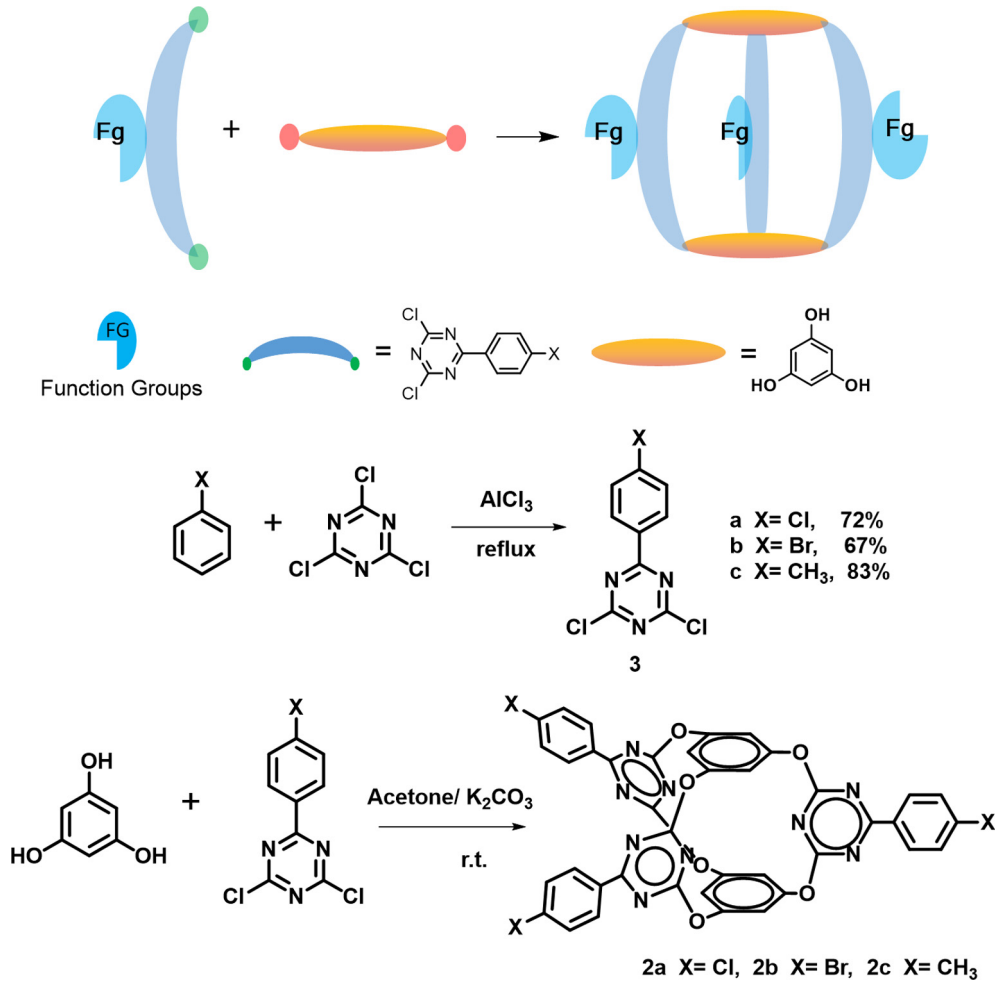

Fig. 2 Synthesis of 2 *via* one-pot synthesis protocol.

Table 2 One-pot synthesis of 2 with different function groups.

3 (X)[a]	Conditions	Yield
X = Cl	2.05 equiv. $Ph(OH)_3$, K_2CO_3, reflux, 4 h	21%
X = Cl	1.05 equiv. $Ph(OH)_3$, K_2CO_3,r.t. 4 h 1.05 equiv. $Ph(OH)_3$,K_2CO_3, reflux 2 h	46%
X = Br	The same as above	43%
X = CH_3	The same as above	48%

a. 3.0 equiv. of **3** was added.

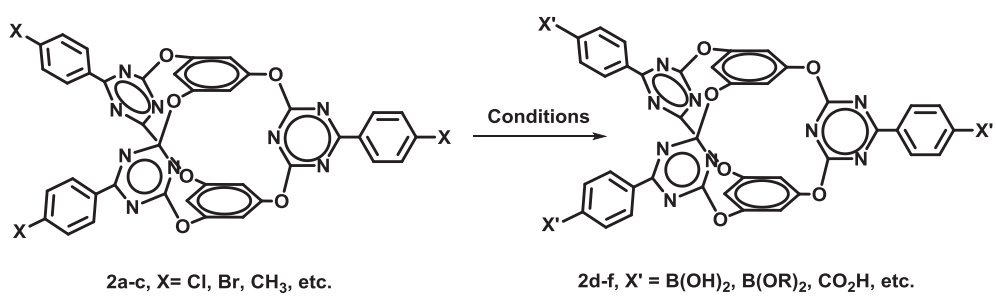

Fig. 3 Synthesis of **2** via one-pot synthesis protocol.

Table 3 Synthesis of other derivatives of **2**.

X	X′	Conditions	Yield
2a	B $(OR)_2$	Dioxane, KOAc, $Pd_2(dba)_3$, PCy_3, 3 h, 80 °C	73%
2d	B $(OH)_2$	Acetone/H_2O, $NaIO_4$, NH_4OAc, 30 °C	94%
2c	CO_2H	CH_2Cl_2, CH_3CN, CrO_3, H_5IO_6, r.t.	76%

The tricarboxylic acid **2f** adopts a slightly twisted D3h symmetric cage conformation. The average length of the bridging C—C bond ($d_{C(benzene)-C(triazine)}$) is 1.48 Å. Two arms of the molecule are straight forward, the other one is slightly twisted (Fig. 4d). **2f** molecules self-assemble in different ways to form similar layers as of **2e** (Fig. 5b), which is around 5.7 Å thick (Figs. S4, S5).

A diamond-shape pore is sandwiched between four **2f** molecules, the size is 11.8 Å * 11.7 Å * 5.7 Å. **2f** molecules interact with each other *via* π-π stacking to form Zigzag chains, and one channel exists between two Zigzag chains, 4.9 Å wide at its narrowest point. The layer staggered to build up the crystal, due to the incompact hollow channels structure, the crystal isn't as stable as **2e**, and it weathers in the air and turns white, which due to the dissociation of solvent occupied the channels.

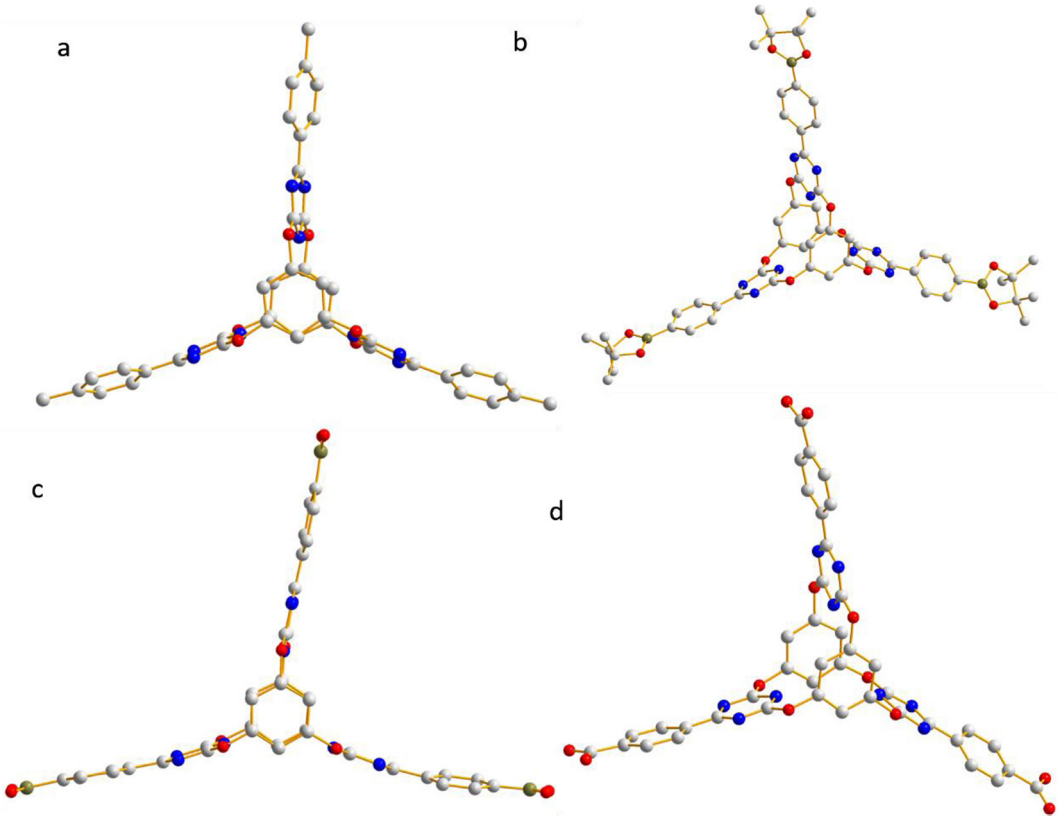

Fig. 4 Top view of X-ray crystal structure of (a) **2c**, (b) **2d**, (c) **2e**, (d) **2f**.

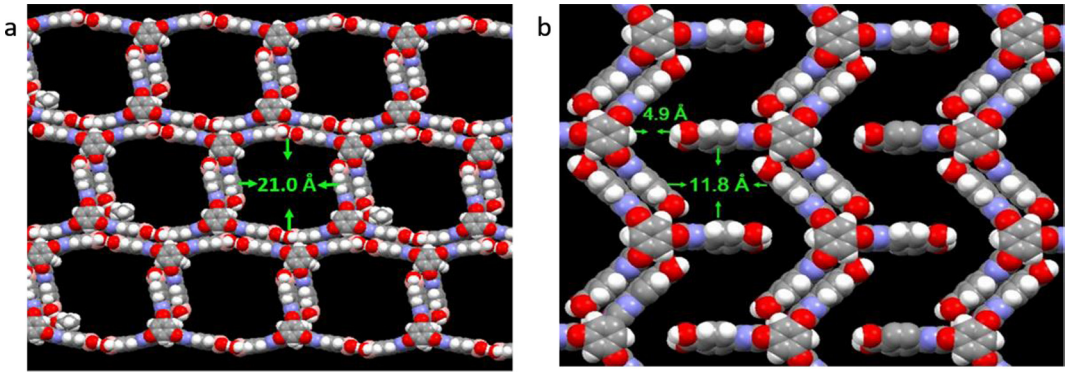

Fig. 5 The 3D honey-comb self-assembly layer of (a) **2e**, (b) **2f** in the solid state.

4. Conclusions

In summary, based on bicyclo(tetraoxacalix[2]arene[2]triazine) scaffold, we designed and prepared a series of bicyclo(tetraox acalix[2]arene[2]triazine) cage compounds, and developed a convenient and practical facile synthesis protocol starting from readily available materials. X-ray diffraction results show these macrocycles adopt D_3h symmetric cage conformations. This brand-new building blocks self-assemble in a distinctive way and form a unique topological structure when hydrogen bonding group exists. Meanwhile, the electron deficient nature of the macrocycles scaffold would endow them the ability for the recognition of anion species. We can expect MOF, COF or other mesoporous materials based on these new building blocks, to be able to recognize specific molecules, which is under investigation in this laboratory now.

Declarations of interest

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Appendix A. Supplementary data

The CCDC No. for compound 2d: CCDC1575504, 2c: CCDC1575505, 2f: CCDC1575506, 2e: CCDC1575507. Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jscs.2017.11.003.

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